

5.05 Hypothesis Tests and Confidence Intervals			
5.05a	The distribution of $\bar{X}$ and the central limit theorem		a) Know that for any randomly and independently selected sample, X_i, of size n taken from a population, then for the sample mean $\bar{X}$: 1. $E(\bar{X}) = \mu$, 2. $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$ and 3. $\bar{X}$ is approximately normally distributed for large n (approximately $n > 25$). [Proof of these results is excluded.]
5.05b	Unbiased estimates of population mean and variance		b) Know that unbiased estimates of the population mean and variance are given by $\frac{\sum x}{n}$ and $\frac{n}{n-1} \left(\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \right)$ respectively. [Proof of these results is excluded.] Only an informal understanding of “unbiased” is required.
5.05c	Using the normal distribution in hypothesis tests		c) Be able to use a normal distribution to carry out a hypothesis test for a population mean in the following cases: 1. a sample drawn from a normal population with known, given or assumed variance, 2. a large sample drawn from any population with known, given or assumed variance, 3. a large sample, drawn from any population with unknown variance.
5.05d	Confidence intervals		d) Be able to use a normal distribution to find a confidence interval for a population mean in the above cases.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
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